

URUMQI DIWOPU INTL, China

WMO: 514635

Lat: 43.907N

Lon: 87.474E

Elev: 648

StdP: 93.78

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-21.9	-26.8	0.4	-22.5	-24.0	0.5	-21.1	4.4	-7.3	3.5	-9.7	0.8	320	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	35.9	18.3	34.2	17.8	33.0	17.5	20.2	30.3	19.4	29.6	18.7	28.9	3.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	13.8	22.2	16.2	12.4	22.4	15.1	11.6	23.3	61.2	30.3	58.2	29.1	55.9	28.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
7.3	5.7	4.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-25.7	39.5	3.0	1.6	-27.9	40.6	-29.6	41.5	-31.3	42.4	-33.5	43.5		
			-24.8	21.7	4.6	1.6	-28.0	22.8	-30.7	23.8	-33.3	24.7	-36.6	25.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-14.2	-10.0	1.5	13.3	19.4	24.6	26.4	25.1	19.4	10.6	0.2	-10.2	(6)
(7)		DBStd	15.07	4.66	4.65	6.85	5.43	4.59	3.62	3.31	3.67	4.63	5.05	5.66	4.86	(7)
(8)		HDD10.0	2594	751	559	273	30	1	0	0	2	56	296	627		(8)
(9)		HDD18.3	4246	1009	793	521	162	43	3	1	2	41	243	544	886	(9)
(10)		CDD10.0	2209	0	0	11	130	292	439	509	469	284	74	1	0	(10)
(11)		CDD18.3	819	0	0	0	12	76	192	251	212	73	3	0	0	(11)
(12)		CDH23.3	8735	0	0	1	102	708	2087	2905	2286	627	19	0	0	(12)
(13)		CDH26.7	3794	0	0	0	21	235	912	1384	1033	206	2	0	0	(13)
(14)	Wind	WSAvg	2.1	1.1	1.3	1.8	2.7	2.8	2.7	2.6	2.5	2.0	1.6	1.2	(14)	
(15)	Precipitation	PrecAvg	232	9	9	13	25	27	37	26	15	15	15	13	(15)	
(16)		PrecMax	310	19	29	30	62	78	62	90	73	44	38	27	32	(16)
(17)		PrecMin	132	2	1	2	1	3	8	4	3	2	1	3	4	(17)
(18)		PrecStd	50	5	6	7	15	17	16	21	19	10	7	7		(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	2.2	21.0	29.8	33.8	36.2	39.0	37.9	34.0	25.8	15.2	2.1	(19)
(20)			MCWB	-1.3	0.0	10.1	13.0	14.9	17.7	18.6	18.1	16.6	12.5	8.5	0.2	(20)
(21)		2%	DB	-3.2	0.0	17.2	26.1	31.1	34.6	36.1	35.2	31.1	22.8	12.1	-0.1	(21)
(22)			MCWB	-4.2	-1.7	8.6	11.7	14.5	17.5	18.7	18.1	15.6	11.7	6.5	-1.1	(22)
(23)		5%	DB	-5.9	-1.9	14.2	23.9	29.2	33.1	34.2	33.9	29.0	20.2	9.8	-1.9	(23)
(24)			MCWB	-6.6	-3.0	7.6	11.3	14.0	17.1	18.4	18.0	14.7	10.6	5.4	-2.7	(24)
(25)		10%	DB	-7.8	-3.8	11.8	21.9	27.2	31.2	32.9	31.9	27.0	18.2	7.8	-3.2	(25)
(26)			MCWB	-8.4	-4.7	6.6	10.3	13.5	16.8	18.0	17.5	14.1	9.7	4.4	-3.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	0.8	10.7	14.4	17.2	20.7	22.3	21.1	18.6	13.5	9.0	0.5	(27)
(28)			MCDWB	-0.4	1.7	18.6	25.1	27.7	32.1	32.6	30.4	29.8	22.9	15.2	1.6	(28)
(29)		2%	WB	-4.4	-1.4	9.1	13.1	16.2	19.3	20.5	19.9	16.7	12.2	7.0	-1.2	(29)
(30)			MCDWB	-3.6	-0.2	16.1	22.4	26.9	29.6	30.4	30.1	27.1	20.6	10.8	-0.4	(30)
(31)		5%	WB	-6.5	-3.1	8.0	12.0	15.4	18.5	19.7	19.1	15.7	11.3	5.6	-2.9	(31)
(32)			MCDWB	-5.9	-2.0	13.9	21.2	25.9	29.0	29.7	29.2	26.6	19.1	9.0	-2.2	(32)
(33)		10%	WB	-8.3	-4.5	6.6	11.2	14.6	17.7	19.0	18.4	14.8	10.1	4.3	-4.1	(33)
(34)			MCDWB	-7.7	-3.6	11.2	19.5	24.7	28.2	29.1	28.5	25.2	17.4	7.3	-3.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	7.7	7.9	10.3	10.6	9.9	10.2	10.3	9.3	6.7	6.7	(35)	
(36)			MCDBR	9.3	9.2	11.3	12.3	12.1	11.7	11.3	11.6	11.7	11.0	9.7	8.0	(36)
(37)			MCWBR	8.2	7.5	5.6	5.1	4.6	4.3	4.3	4.4	4.8	5.2	5.8	6.7	(37)
(38)		5% WB	MCDWR	9.1	8.6	10.6	11.0	10.7	10.0	9.8	10.2	10.6	10.4	9.1	7.7	(38)
(39)			MCWBR	8.0	7.1	5.7	4.9	4.6	4.0	4.1	4.2	4.6	5.3	5.7	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.348	0.401	0.458	0.513	0.500	0.510	0.490	0.470	0.445	0.430	0.386	0.350	(40)	
(41)		taud	2.064	1.973	1.903	1.826	1.900	1.937	2.030	2.066	2.055	2.020	2.077	2.106	(41)	
(42)		Ebn at Noon	730	753	762	755	782	774	785	786	768	712	678	686	(42)	
(43)		Edh at Noon	106	139	170	199	190	184	166	155	145	131	104	93	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.95	2.87	4.19	5.53	6.45	6.59	6.44	5.89	5.02	3.52	2.15	1.57	(44)	
(45)		RadStd	0.09	0.17	0.27	0.34	0.32	0.32	0.29	0.26	0.16	0.19	0.18	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.69	N/A	N/A	+131	+90

Nomenclature: See separate page